SIMATS ENGINEERING

SIMATS Online Programming Lab

JAVA PROGRAMMING

Kishore Kumar V
192521371RD

QUESTIONS

- **Q1** Palindrome number/string — 10 marks
- **Q2** First n perfect numbers — 10 marks
- **Q3** Decimal number equivalent to binary number and octal number — 10 marks
- **Q4** Nth max number and nth min number in array then sum of it and difference of it — 10 marks
- **Q5** Find the average positive and negative numbers upto -1 — 10 marks
- **Q6** Removing duplicates in array — 10 marks
- **Q7** Matrix multiplication — 10 marks

Q4 Nth max number and nth min number in array then sum of it and difference of it

10 marks

Nth max number and nth min number in array then sum of it and difference of it

Your Code

```java
import java.util.Arrays;
import java.util.Scanner;
public class NthMaxMin {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        System.out.print("Enter array size: ");
        int n = sc.nextInt();
        int[] arr = new int[n];
        System.out.println("Enter array elements:");
        for (int i = 0; i < n; i++) {
            arr[i] = sc.nextInt();
        }
        Arrays.sort(arr);
        System.out.print("Enter N: ");
        int k = sc.nextInt();
        int nthMin = arr[k - 1];
        int nthMax = arr[n - k];
        int sum = nthMin + nthMax;
        int diff = nthMax - nthMin;
        System.out.println("Nth Minimum = " + nthMin);
        System.out.println("Nth Maximum = " + nthMax);
        System.out.println("Sum = " + sum);
        System.out.println("Difference = " + diff);
        sc.close();
```

Input

```
5
10 20 30 40 50
3
```

Output

```
Enter array size: Enter array elements:
Enter N: Nth Minimum = 30
Nth Maximum = 30
Sum = 60
Difference = 0
```